

CENK CELIK

Research Fellow at University College London

cenk.celik@ucl.ac.uk • cenk.celik@proton.me • ORCID: 0000-0001-8301-0172 • Google Scholar • London WC1E 6BT

Research Fellow and computational biologist specialising in cell state transitions and tumour microenvironments, with a focus on cell cycle arrest and EMT in cancer. Strong background in experimental biology and expertise in spatial and single-cell transcriptomics to study tumour heterogeneity and immune interactions. Independent project leader with a collaborative, transparent and rigorous approach. Trained in Singapore and the UK. Proven track record in high-impact publications, method development and postgraduate supervision. Committed to translational cancer research through open, interdisciplinary science.

Professional Experience

Research Fellow - University College London, UK

Jun 2023 – present

- Developing analytical framework for spatial -omics biology
- Investigating Go arrest in breast cancer using bulk, single-cell and spatial transcriptomics datasets
- Researching cell states such as EMT using cutting-edge datasets
- Mentoring master's and PhD students with one-on-one weekly meetings

Research Fellow - Nanyang Technological University, Singapore

Mar 2021 – May 2023

- Led a project researching *in vivo* and *in vitro* host-pathogen synergy in wound infection
- Conducted and analysed next-generation sequencing data by developing pipelines
- Supported and supervised undergraduate and PhD students
- Used *C. elegans* and mouse models for different projects focusing on pathologies of endoplasmic reticulum stress
- Drafted, edited, revised and contributed data to papers for submission in high-impact journals (2/7 first author)
- Trained new research staff/students

Research Assistant - National University Hospital System, Singapore

Nov 2019 – Aug 2020

- Produced and analysed data to create representative graphs highlighting results for presentations
- Planned, modified and executed research techniques, procedures and tests for nano-fibrous scaffold fabrication
- Performed statistical, qualitative and quantitative analysis
- Mentored new research staff

Graduate Assistant - National University of Singapore, Singapore

Jan 2016 – May 2018

- Instructed undergraduates on the fundamentals of cell culture and methods for collecting, analysing and reporting data
- Mentored master's students throughout all phases of their research projects

Education

PhD in Medicine - National University of Singapore, Singapore

Aug 2015 – Aug 2020

- Researched scaffold-mediated stem cell chondrogenesis by pulsed electromagnetic fields
- Published a first-author research article (IF = 10.633) and contributed to another research article (IF = 8.079)
- Presented at two international conferences
- Awarded Singapore International Graduate Award scholarship by NUS
- Awarded Young Orthopaedic Researcher Award by APOA Sports Meeting '19
- Awarded NUS Researcher Travel Grant by the National University of Singapore
- Wrote a first-author book chapter

MSc in Bioengineering - Ege University, Turkey

Jul 2012 – May 2015

- Produced and characterised hybrid tissue scaffold approaches for 3D vascularised bone tissue
- Awarded TUBITAK 1002 and TUBITAK Career Grant scholarships

- Received Republic of Turkey Prime Minister Scholarship (4 years)
- Awarded and administered TUBITAK 2209 University Students Research Projects Support Program

Publications

Research articles

- Tan AMZ, **Celik C**, Lee SYT, Veleba M, Manzano C, Abdul RMK, Thibault G, Kline KA (2025). Enterococcus faecalis redox metabolism activates the unfolded protein response to impair wound healing. *bioRxiv*. [doi: [10.1101/2025.08.07.669218](https://doi.org/10.1101/2025.08.07.669218)].
- **Celik C*** & Secrier M (2025). EnrichMap: Spatially informed enrichment analysis for functional interpretation of spatial transcriptomics. *bioRxiv*. [doi: [10.1101/2025.05.30.656960](https://doi.org/10.1101/2025.05.30.656960)]. *corresponding author
- da Silva RAG, Tien BYQ, Hsien PNK, **Celik C**, Tan AZC, Ismail MH, Hu G, Chong KKL, Thibault G, Chen J, Kline KA (2025). *Enterococcus faecalis*-derived lactic acid facilitates persistent and polymicrobial wound infections by suppressing macrophage activation. *bioRxiv*. [doi: [10.1101/2025.01.31.635924](https://doi.org/10.1101/2025.01.31.635924)]
- **Celik C**, Withnell E, Pan S, Chu T, Labbadia J, Secrier M (2025). Balancing tumour proliferation and sustained cell cycle arrest through proteostasis remodelling drives immune niche compartmentalisation in breast cancer. *bioRxiv*. [doi: [10.1101/2025.01.08.632014](https://doi.org/10.1101/2025.01.08.632014)]
- **Celik C**, Lee SYT, Tanoto FR, Veleba M, Kline KA, Thibault G (2024). Decoding complexity of delayed wound healing following *Enterococcus faecalis* infection. *eLife* 13:RP95113. [doi: [10.7554/eLife.95113.3](https://doi.org/10.7554/eLife.95113.3)]
- Kaliya-Perumal AK, **Celik C**, Carney TJ, Harris MP, Ingham PW (2024). Genetic regulation of injury induced heterotopic ossification in adult zebrafish. *Dis Model Mech* dmm.050724. [doi: [10.1242/dmm.050724](https://doi.org/10.1242/dmm.050724)]
- Dudkevich R, Koh JH, Beaudoin-Chabot C, **Celik C**, Lebenthal-Loinger S, Karako-Lampert S, Thibault G, Henis-Korenblit S (2022). Neuronal IRE-1 coordinates an organism-wide cold-stress response by regulating fat metabolism. *Cell Reports* 41, 111739. [doi: [10.1016/j.celrep.2022.111739](https://doi.org/10.1016/j.celrep.2022.111739)]
- Beaudoin-Chabot C, Wang L, **Celik C**, Abdul-Khalid ATF, Thalappilly S, Xu S, Koh JH, Lim VWX, Low AD, Thibault G (2022). The unfolded protein response reverses the effects of glucose on lifespan in chemically-sterilised *C. elegans*. *Nat Commun* 13, 5889. [doi: [10.1038/s41467-022-33630-0](https://doi.org/10.1038/s41467-022-33630-0)]
- Yang Z, **Celik C**, Parate D, Franco-Obregón A, Lee EH (2022). The application of pulsed electromagnetic field for cartilage regeneration. *Tissue Eng Part A* 28, S-365. [doi: [10.1089/ten.tea.2022.29025.abstracts](https://doi.org/10.1089/ten.tea.2022.29025.abstracts)]
- **Celik C**, Franco-Obregón A, Hui JHP, Lee EH & Yang Z (2021). Directionalities of magnetic fields and topographic scaffolds synergise to enhance MSC chondrogenesis. *Acta Biomater* 119, 169-183. [doi: [10.1016/j.actbio.2020.10.039](https://doi.org/10.1016/j.actbio.2020.10.039)]
- Parate D, Kadir ND, **Celik C**, Lee EH, Hui JHP, Franco-Obregón A & Yang Z (2020). Pulsed electromagnetic fields potentiate the paracrine function of mesenchymal stem cells for cartilage regeneration. *Stem Cell Res & Ther* 11:46. [doi: [10.1186/s13287-020-1566-5](https://doi.org/10.1186/s13287-020-1566-5)]
- Cetmi SD, Renkler NZ, Kose A, **Celik C** & Oncel SS (2019). Preparation of electrospun polycaprolactone nanofiber mats loaded with microalgal extracts. *Eng Life Sci* 19:691-699. [doi: [10.1002/elsc.201900009](https://doi.org/10.1002/elsc.201900009)]
- Duman OM, **Celik C**, Sarıkanat M & Sendemir A (2014). Investigation of central nervous system neurons under mechanical strain: An *in vitro* traumatic brain injury model. *18th National Biomedical Engineering Meeting*, 1-4. [doi: [10.1109/biyomut.2014.7026384](https://doi.org/10.1109/biyomut.2014.7026384)]

Reviews

- **Celik C**, Pan S, Withnell E, Lam HW, Secrier M (2025). Decrypting cancer's spatial code: from single cells to tissue niches. *Mol Oncol*. [doi: [10.1002/1878-0261.70100](https://doi.org/10.1002/1878-0261.70100)]
- **Celik C**, Lee SYT, Yap WS, Thibault G (2023). Endoplasmic reticulum stress and lipids in health and disease. *Progress in Lipid Research* 8, 101198. [doi: [10.1016/j.plipres.2022.101198](https://doi.org/10.1016/j.plipres.2022.101198)]

Book chapters

- **Celik C**, Mogal VT, Hui JHP, Loh XJ & Toh WS (2018). Injectable Hydrogels for Cartilage Regeneration. In: Thakur V, Thakur M (eds) *Hydrogels. Gels Horizons: From Science to Smart Materials*. Springer, Singapore. [doi: [10.1007/978-981-10-6077-9_12](https://doi.org/10.1007/978-981-10-6077-9_12)]

- Karaman O, **Celik C** & Sendemir A (2018). Self-Assembled Biomimetic Scaffolds for Bone Tissue Engineering. In Management Association, I. (Ed.), *Biomedical Engineering: Concepts, Methodologies, Tools and Applications*, 476-504, Hershey, PA: IGI Global. [[doi: 10.4018/978-1-5225-3158-6.ch021](https://doi.org/10.4018/978-1-5225-3158-6.ch021)]

Invited talks

- **Celik C** (May 10, 2025). Spatial evolution of proliferative and arrested niches in breast cancer. *AI in Pharma*, Hacettepe University, Ankara, Türkiye.
- **Celik C** (Sep 23, 2024). Internal and external regulation of cell cycle arrest in breast cancer. *Izmir Biomedicine and Genome Centre*, Izmir, Türkiye.

Proceedings

- **Celik C**, Withnell E, Secrier M (Oct 7, 2024). Intrinsic and extrinsic factors influencing G0 cell cycle arrest in breast cancer [Selected talk]. *Crick Cancer Research Symposium 2024*, London, UK.
- **Celik C**, Withnell E, Secrier M (May 1, 2024). Unravelling internal and external regulatory mechanisms of cell cycle arrest in breast cancer [Poster]. *UCL Cancer Symposium 2024*, London, UK.
- **Celik C**, Secrier M (Jan 24-25, 2024). Deciphering the interplay between cell cycle arrest and tumour microenvironment in breast cancer [Selected talk & Poster]. *Festival of Genomics & Biodata*, London, UK.
- **Celik C**, Secrier M (Oct 8-10, 2023). Deciphering the interplay between cell cycle arrest and tumour microenvironment in breast cancer: Insights from single-cell data [Poster]. *4th Crick International Cancer Conference*, London, UK.
- Lee STL, **Celik C**, Tan AMZ, Veleba M, Kline KA, Thibault G (Dec 3-7, 2022). Dissecting the synergistic role of the unfolded protein response in wound infections [Poster & Oral]. *ACSB & EMBO Meeting*, Washington DC, USA.
- **Celik C**, Yang Z, Franco-Obregón A, Hui JHP (Apr 4-7, 2019). Altering MSC chondrogenesis by PEMF [Oral]. *APOA Sports Meeting 19*, Kuala Lumpur, Malaysia.
- **Celik C**, Yang Z, Franco-Obregón A, Hui JHP (Sep 9-11, 2018). Enhancement of MSC chondrogenesis by PEMF [Poster]. *5th TERMIS World Congress*, Kyoto, Japan.
- Cetmi SD, Renkler NZ, Kose A, **Celik C**, Sendemir A & Oncel SS (Nov 11, 2015). Utilisation of microalgal extracts for construction of tissue scaffolds with electrospinning techniques [Poster]. *7th International Bioengineering Congress*, Izmir, Türkiye.
- Duman OM, Tasdemir S, Minaz MC, **Celik C** & Sendemir A (Nov 11, 2013). Fe₃O₄ reinforced polycaprolactone nanofibrous scaffolds [Poster]. *VIth International Bioengineering Congress: Human Welfare*, Kusadasi, Türkiye.
- **Celik C**, Karaman O, Sendemir A (Nov 11, 2013). More than a picture: Biomedical Illustration [Oral]. *19th International Biomedical Science and Technology Symposium*, Kusadasi, Türkiye.
- **Celik C**, Gorgun C, Sendemir A (Nov 11, 2013). Cell viability of keratinocytes on electrospun Spirulina/PCL composites [Poster]. *19th International Biomedical Science and Technology Symposium*, Kusadasi, Türkiye.
- Duman OM, Cogan S, Minaz MC, **Celik C** & Sendemir A (Sep 9, 2013). PCL/Fe₃O₄ Scaffold Production with Electrospinning [Oral]. *Advanced Materials World Congress 2013*, Cesme, Türkiye.
- Serdengeçti C, Sen I, Duman OM, **Celik C**, Sendemir A & Seki Y (Apr 4, 2013). Production and analysis of parameters affect morphological characters of chitosan nanospheres [Poster]. *1st Ege University Nanotechnology Days*, Izmir, Türkiye.
- Duman OM, **Celik C**, Minaz MC, Sendemir A (Apr 4, 2013). Fabrication of PEO/Fe₃O₄ composite tissue scaffolds by electrospinning [Poster]. *1st Ege University Nanotechnology Days*, Izmir, Türkiye.
- Demirkaya C, **Celik C**, Sarikanat M, Conk-Dalay M, Sendemir A (May 5, 2012). Applications of microalgae for tissue engineering [Oral]. *Marine Biotechnology and Genomics Workshop*, Bodrum, Türkiye.
- **Celik C**, Demirkaya C, Sarikanat M, Conk-Dalay M & Sendemir A (Nov 11, 2011). Effects of microalgae reinforcement on biocompatibility and mechanical properties of tissue engineering scaffolds [Oral]. *1st National Aegean Composite Materials Symposium*, Selcuk, Türkiye.

Teaching

- Life at the single-cell and spatial scale. *Bootcamp* (Aug 2025) at Bilimler Köyü, Foça, Türkiye.
- Art of ATGC: Genetic data analysis and visualisation. *Bootcamp* (Aug 2024) at Bilimler Köyü, Foça, Türkiye.
- Graduate Teaching Assistantship Programme (2016 - 2018). National University of Singapore.

Proficiencies

- *R/RStudio* • Python 3 • *UNIX/Bash* • Cell Ranger • Loupe Browser • NGS data analysis (bulk, single-cell RNA-seq & spatial transcriptomics)
- Documentation/Version control (GitHub, .Rmd, ReadTheDocs) • Image processing • Affinity Designer/Photo

Core Competencies

- Single-cell & bulk RNA-seq • qPCR • Western blot • Cell culture • Confocal imaging • Immunofluorescence
- Handling of mice & rats • Lentiviral transfection/transduction • Primer design • Single-cell isolation • Flow cytometry

Memberships

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| - The European Association for Cancer Research - Member | Feb 2024 – present |
| - The Genetics Society UK - Member | Aug 2023 – Aug 2024 |
| - Society for Cell Biology Singapore - Member | Sep 2022 – May 2023 |
| - DIE at NTU School of Biological Sciences - Member | Mar 2022 – May 2023 |

Certifications

- **Responsible Care & Use of Laboratory Animals** - SingHealth, Apr 2021, ID: SEMC/RCULAC/2021/040
- **Genomic Data Science Specialisation** - John Hopkins University, Sep 2020, ID: D7S7D536DXDD
- **Data Analysis for Life Science** - HarvardX, Aug 2020, ID: a1cb1feb86a24a6b8d28873249cf1bad
- **Python Data Science** - edX & IBM, Jun 2020, ID: 8e35eee7e19b4071a2a1109e599b8800

Awards & Honours

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| - Crick Cancer Symposium 2024 - Selected talk, London | Oct 2024 |
| - Poster Presentation Winner - Festival of Genomics & Biodata, London | Jan 2024 |
| - Translational Researcher Programme - 10X Genomics, <i>Inc.</i> , Singapore | Feb 2022 |
| - Young Orthopaedic Researcher - APOA Sports Meeting '19, Kuala Lumpur | Apr 2019 |
| - Researcher Travel Grant - National University of Singapore, Singapore | Sep 2018 |
| - Singapore International Graduate Award - National University Singapore, Singapore | Aug 2015 – Aug 2019 |
| - Research Scholarship - TUBITAK 1002, Turkey | Jun 2014 – May 2015 |
| - Research Scholarship - TUBITAK Career Grant, Turkey | Sep 2012 – Sep 2013 |
| - University Students' Research Project Support Program - TUBITAK 2209, Turkey | Sep 2011 – Jan 2012 |
| - Prime Minister Scholarship - General Directorate of Credit & Dormitories Agency, Turkey | Jan 2008 – Jun 2012 |

Volunteering

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| - Life at the single-cell and spatial scale (bootcamp) - Bilimler Köyü | 11–17 Aug 2025 |
| - Computational Research Exchange Cancer Network - UCL | Dec 2024 – Mar 2025 |
| - Art of ATGC: Genetic Data Analysis and Visualisation (bootcamp) - Bilimler Köyü | 12–18 Aug 2024 |
| - Bioinformatics Instructor (bootcamp) - KizCode | Oct 2021 – Jan 2024 |
| - Executive Committee Member - Singapore-Turkey Friendship Association | May 2016 – Jun 2018 |
| - Organising Committee - 19 th International Biomedical Science & Technology Symposium | Nov 2012 – Dec 2012 |
| - Organising Committee - 6 th International Bioengineering Congress | Nov 2012 – Dec 2012 |